• Unit 3 • AOS 1 Data Analysis • Lesson 2

What shape is your data?

1C Displaying and describing numerical data · 1D Dot plots and stem plots · 1E Log scales

Do Now | APK | Learn | Worked examples | You try | Exit tasks

TODAY'S LESSON

Today's lesson

LEARNING INTENTIONS

- [ ] To be able to construct frequency tables for discrete numerical data.
- [ ] To be able to construct frequency tables for grouped numerical data (discrete and continuous).
- [ ] To be able to construct a histogram from a frequency table for numerical data.
- [ ] To be able to construct a histogram from numerical data using a CAS calculator.
- [ ] To be able to describe the distribution of numerical data according to its key characteristics of shape, centre, spread and outliers.
- [ ] To be able to construct a dot plot for numerical data.
- [ ] To be able to construct a stem plot for numerical data.
- [ ] To be able to revise the concept of $\log_{10} x$.
- [ ] To be able to investigate the effect of the logarithmic scale on the features of a distribution.

LESSON SEQUENCE

1. Do Now · three current VCAA questions
2. APK · Year 11 skills
3. 1C · frequency tables → WE 6, 7
4. 1C · histograms → WE 8 · CAS
5. 1C · shape, centre, spread, outliers → WE 9
6. 1D · dot plots and stem plots → WE 10, 11
7. 1E · logs and log scales → WE 12, 13
8. Exit tasks · three VCAA questions
9. Check-in and homework

SUCCESS CRITERIA

By the end of this lesson you'll be able to say...

- ☐ I can construct a frequency table for discrete numerical data.
- ☐ I can construct a grouped frequency table.
- ☐ I can construct a histogram from a grouped frequency table.
- ☐ I can describe the features of a distribution from a histogram.
- ☐ I can construct a dot plot.
- ☐ I can construct a stem plot.
- ☐ I can use a CAS calculator to find logs.
- ☐ I can interpret a histogram with a log scale.

DO NOW 1 OF 3 · VCAA 2024 NHT · EXAM 1

VCAA: NHT report gives answers only

Review of 1A

The variables *exercise type* (aerobic, boxing, circuit) and *recovery time* (short, medium, long) are

A. a nominal variable and a numerical variable respectively.

B. a nominal variable and an ordinal variable respectively.

C. an ordinal variable and a nominal variable respectively.

D. both ordinal variables.

E. both nominal variables.

SOLUTION · B

exercise type only names the exercise: **nominal**. *recovery time* (short, medium, long) has a natural order: **ordinal**.

WHAT'S THE DISTRACTOR?

No % data: the NHT report gives answers only. Watch for C (the order swapped).

Multiple choice

DO NOW 2 OF 3 · VCAA 2023 · EXAM 2

VCAA: 50% scored 1/1

Data was collected to investigate the use of electronic images to automate the sizing of oysters for sale.
The variables in this study were:

- *ID*: identity number of the oyster
- *weight*: weight of the oyster in grams (g)
- *volume*: volume of the oyster in cubic centimetres (cm^3)
- *image size*: oyster size determined from its electronic image (in megapixels)
- *size*: oyster size when offered for sale: small, medium or large

The data collected for a sample of 15 oysters is displayed in Table 1.

a. Write down the number of categorical variables in Table 1. (1 mark)

SOLUTION

2: *ID* and *size*. ID numbers are labels, not quantities.
Examiner: a number of students didn't answer; no need to name the variables.

Table 1

ID	Weight (g)	Volume (cm^3)	Image size (megapixels)	Size
1	12.9	13.0	5.1	large
2	11.4	11.7	4.8	medium
3	17.4	17.4	6.5	large
4	6.8	7.2	2.9	small
5	9.6	10.1	3.7	medium
6	15.5	15.6	5.7	large
7	9.7	9.9	4.0	small
8	7.0	7.5	2.7	small
9	12.6	12.7	5.5	medium
10	12.5	12.7	5.0	medium
11	10.1	10.5	3.9	medium
12	10.6	10.8	4.1	medium
13	13.0	13.1	5.3	large
14	8.1	8.5	3.5	small
15	14.1	14.2	5.3	large

Quick write

DO NOW 3 OF 3 · VCAA 2023 · EXAM 2

VCAA: **95%** scored 1/1

The following data shows the sizes of a sample of 20 oysters rated as small, medium or large.

small	small	large	medium	medium
medium	large	small	medium	medium
small	medium	small	small	medium
medium	medium	medium	small	large

i. Use the data above to complete the following frequency table. (1 mark)

Table 2

Size	Frequency	
	Number	Percentage (%)
small	7	35
medium	10	50
large	3	15
Total	**20**	**100**

EXAMINER REPORT

Well done. Some students left out the total or wrote tally marks instead of numbers: not accepted.

1. **Count**
   small 7, medium 10, large 3
2. **Number**
   Fill the number column; total 20.
3. **Percent**
   $\frac{7}{20} \times 100 = 35\%$
4. **Check**
   $35 + 50 + 15 = 100$

Quick write

APK · FROM YEAR 11

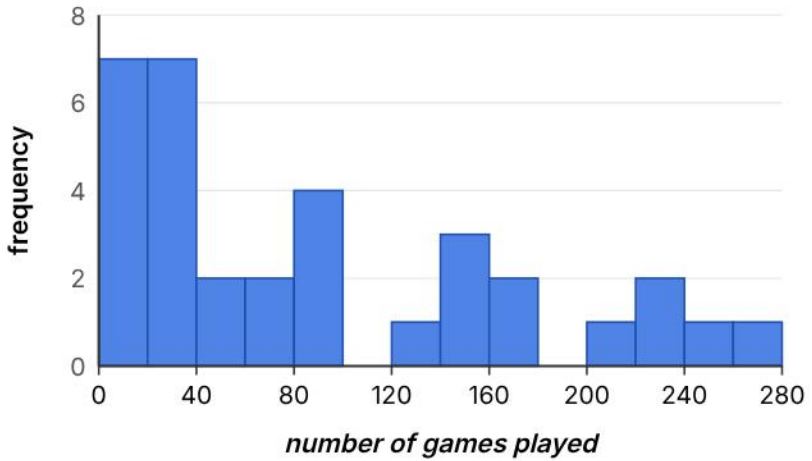
What do you remember?

The following data gives the number of games played in total by each member of an AFL club:

266 259 238 227 210 160 160 159 155 145 133
99 91 80 80 75 73 58 42 36 32 25
32 25 25 21 17 13 10 9 9 4 2

Use a CAS calculator to construct a histogram starting at 0 with column width 20.

Describe the shape of the histogram of the number of games played (above).



Positively skewed: most players have played fewer than 100 games; the tail runs out to the right.

Quick write

APK · FROM YEAR 11

And a stem plot

Construct a stem plot of the ages of a group of people at a party:

19 21 22 56 34 19 27 28 34 17
66 37 20 20 21 45 26 19 23 29

Unordered

age (years) Key: 1 | 7 = 17

1	9 9 7 9
2	1 2 7 8 0 0 1 6 3 9
3	4 4 7
4	5
5	6
6	6

Ordered

age (years) Key: 1 | 7 = 17

1	7 9 9 9
2	0 0 1 1 2 3 6 7 8 9
3	4 4 7
4	5
5	6
6	6

Quick write

APK · FROM YEAR 11

Logs and powers of ten

a Find the log of 592 to two decimal places.

b What is the order of magnitude of 478 000?

a $\log_{10}(592) = 2.772\ldots \rightarrow$ **2.77**

b $478\,000 = 4.78 \times 10^5 \rightarrow$ order of magnitude **5**

Quick write

LEARN · 1C

Frequency tables for numerical data

DISCRETE, FEW VALUES

Same as categorical data: one row for **every value** from the minimum to the maximum.

e.g. number of bedrooms: 1, 2, 3, 4, 5

MANY VALUES, OR CONTINUOUS

Group the data into intervals: a **grouped frequency table**.

- every value in an interval
- no overlaps, no gaps
- about 5 to 15 intervals
- easy widths: 10, 100, 1000...
- start each interval on an exact value: 0–49, 50–99

Read it as: 30–<40 means at least 30 but less than 40.

Class discussion

WORKED EXAMPLE 6 · 1C

▶ Video

Constructing a frequency table for discrete numerical data taking a small number of values

The number of bedrooms in each of the 24 properties sold in a certain area over a one month period are as follows:

2 3 4 3 3 4 3 4 4 1 3 2 1 2 2 2 4 5 3 4
4 5 3 4

Construct a table for these data showing both frequency and percentage frequency, giving the values of the percentage frequency rounded to one decimal place.

Number of bedrooms	Frequency	
	Number	%
1	2	8.3
2	5	20.8
3	7	29.2
4	8	33.3
5	2	8.3
Total	**24**	**99.9**

1 Min and max
Minimum 1, maximum 5.

2 Set up
One row for every value, 1 to 5.

3 Count
Count the 1s, 2s, ...; total 24.

4 Percent
$\frac{2}{24} \times 100 = 8.3\%$

5 Total
99.9%
Rounding: that's fine.

Cold calling

WORKED EXAMPLE 7 · 1C

Video

Constructing a grouped frequency table

The data below give the average hours worked per week in 23 countries.

35.0 48.0 45.0 43.0 38.2 50.0 39.8 40.7 40.0 50.0
35.4 38.8 40.2 45.0 45.0 40.0 43.0 48.8 43.3 53.1
35.6 44.1 34.8

Construct a grouped frequency table with five intervals.

Average hours worked	Frequency	
	Number	Percentage
30.0–34.9	1	4.3
35.0–39.9	6	26.1
40.0–44.9	8	34.8
45.0–49.9	5	21.7
50.0–54.9	3	13.0
Total	**23**	**99.9**

1 **Set up**
Five intervals: 30.0–34.9, ..., 50.0–54.9.

2 **List**
Intervals in ascending order.

3 **Count**
Count the countries in each interval.

4 **Percent**
$\frac{1}{23} \times 100 = 4.3\%$

5 **Total**
Total both columns.

Cold calling

YOU TRY · 1C

Constructing a grouped frequency table

The height (cm) of 20 plants in a Year 6 science experiment were recorded.

32.0 40.2 40.5 45.1 47.0 49.1 50.1 53.7 54.2 55.3
56.9 57.2 58.2 67.2 68.9 69.0 72.3 77.6 82.1 88.5

Construct a grouped frequency table with intervals of 10.

height (cm)	frequency	
	number	%
30–<40	1	5
40–<50	5	25
50–<60	7	35
60–<70	3	15
70–<80	2	10
80–<90	2	10
total	**20**	**100**

1 **How many intervals?**
Min 32.0 → start at 30. Max 88.5 → end at 90. Six intervals.

2 **Set up**
Each interval includes the lower bound, not the upper: 30–<40.

3 **Count**
Count the values in each interval; total 20.

4 **Percent**
$\frac{\text{frequency}}{\text{total frequency}} \times 100$

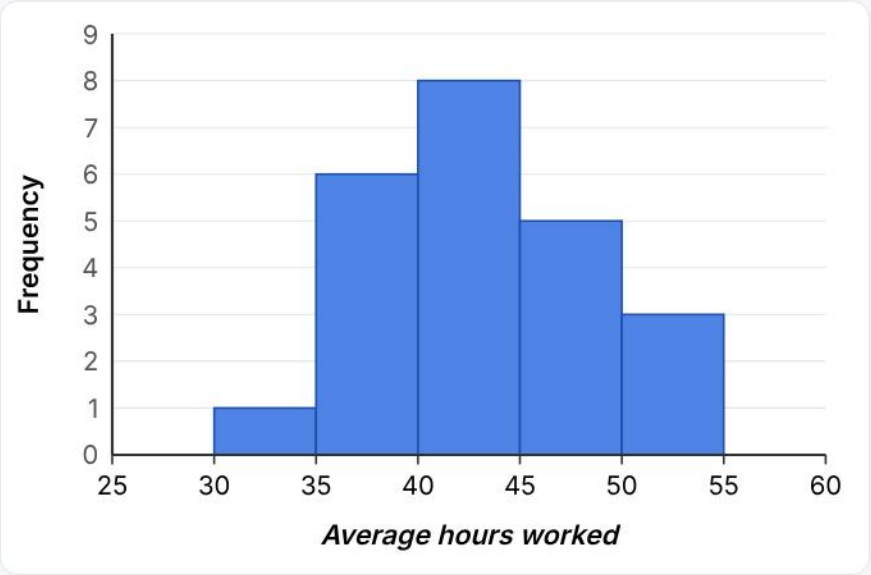
Quick write

LEARN · 1C

The histogram

The graph of a frequency table for a **numerical** variable.

- frequency (count or %) on the vertical axis
- the variable on the horizontal axis
- each bar is one interval; its height is the frequency
- **bars touch**: the scale is continuous



Class discussion

WORKED EXAMPLE 8 · 1C

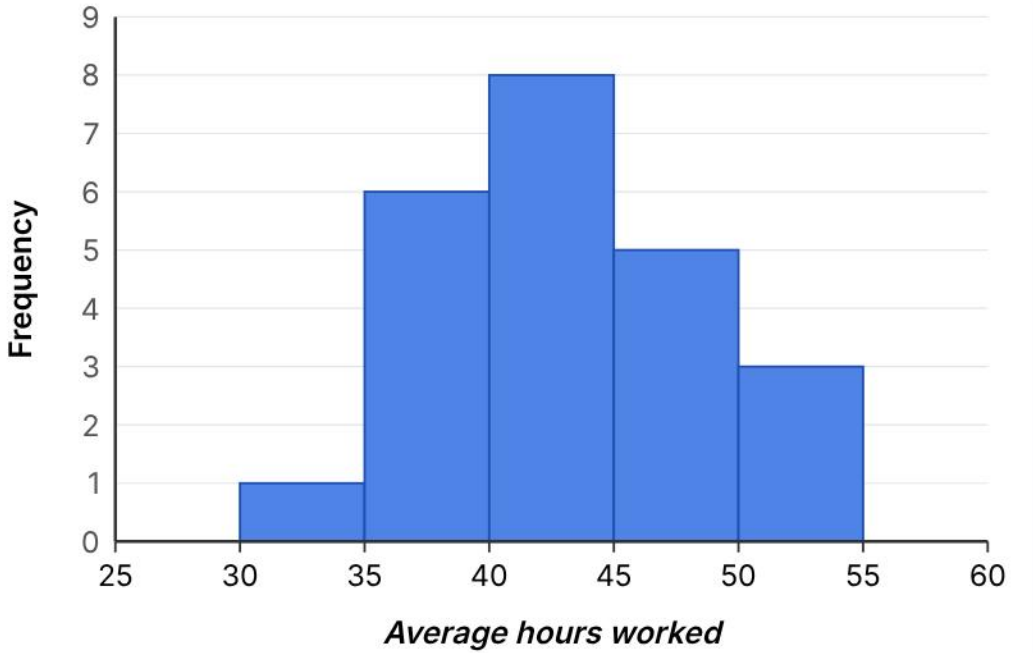
Video

Constructing a histogram from a frequency table

Construct a histogram for the frequency table opposite.

Average hours worked	Frequency
30.0–34.9	1
35.0–39.9	6
40.0–44.9	8
45.0–49.9	5
50.0–54.9	3
Total	**23**

✏ **Use a ruler.** Bars touch; mark the scale at the start of each interval.



1 Label the horizontal axis, mark the start of each interval · 2 Label the vertical axis, scale to the maximum (8) · 3 Draw each bar to its frequency.

Cold calling

YOU TRY · 1C

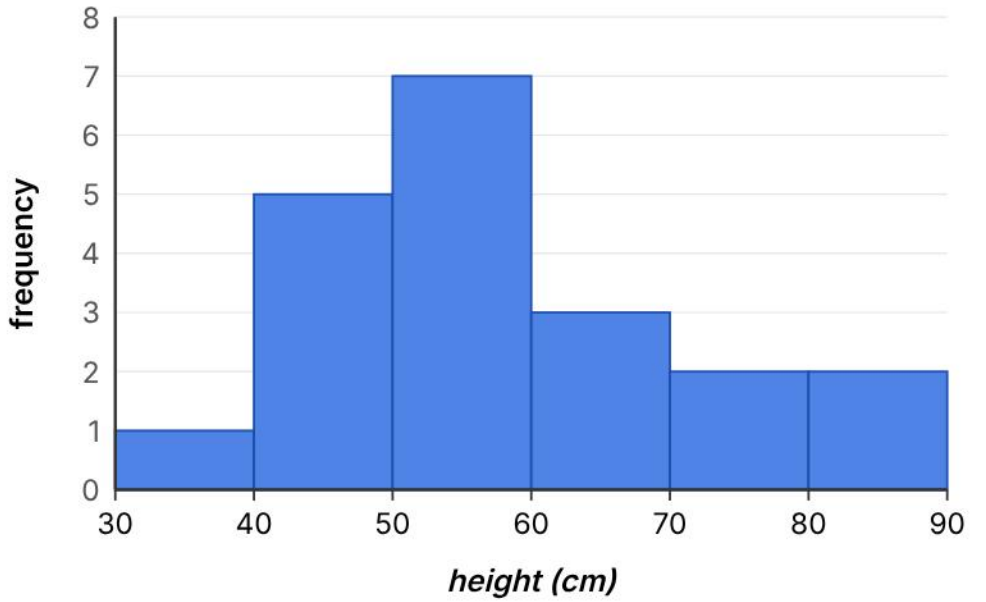
Constructing a histogram from a frequency table

The height (cm) of 20 plants in a Year 6 science experiment was recorded in the following grouped frequency table.

a. Construct a histogram to display this data.

b. What percentage of plants were shorter than 60 cm?

height (cm)	frequency	
	number	%
30–<40	1	5
40–<50	5	25
50–<60	7	35
60–<70	3	15
70–<80	2	10
80–<90	2	10
total	**20**	**100**



b $1 + 5 + 7 = 13$, $\frac{13}{20} = 0.65 \rightarrow$ **65%**

Quick write

CAS · 1C

Histogram from raw data on CAS

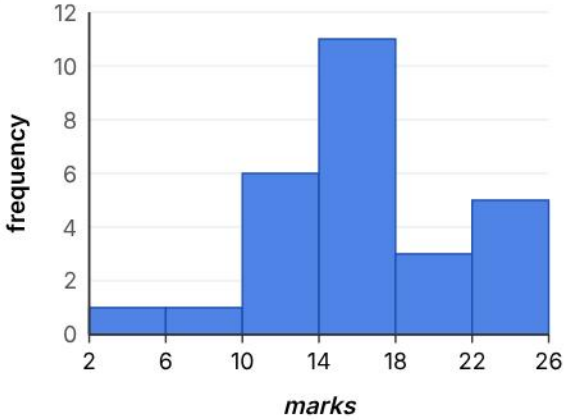
Display the following set of 27 marks in the form of a histogram (column width 4, starting point 2): 16 11 4 25 15 7 14 13 14 12 15 13 16 14 15 12 18 22 17 18 23 15 13 17 18 22 23

TI-NSPIRE

1. New document → **Lists & Spreadsheet**. Name column A **marks**; enter the data.
2. **Add Data & Statistics**; put **marks** on the x-axis.
3. Menu › Plot Type › **Histogram**.
4. Plot Properties › Histogram Properties › Bin Settings › **Equal Bin Width**: Width 4, Alignment 2.
5. Window/Zoom › **Zoom-Data**.

CLASSPAD

1. **Statistics** app. Name list1 **marks**; enter the data.
2. SetGraph: Draw On, Type **Histogram**, XList main\marks, Freq 1. Set.
3. Tap the graph icon.
4. Set Interval: **HStart 2**, **HStep 4**. OK.
5. Analysis › Trace to read each column.

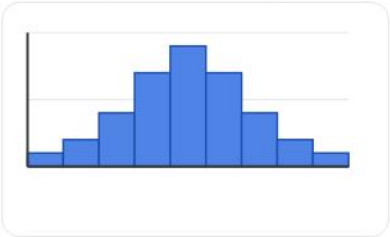


Class discussion

LEARN · 1C

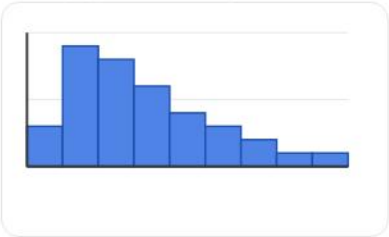
Shape: what to look for

SYMMETRIC



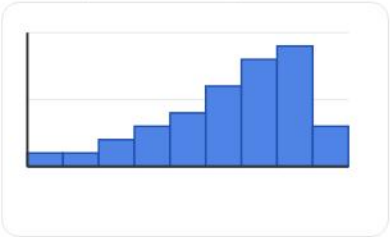
tails off evenly either side of the peak

POSITIVELY SKEWED



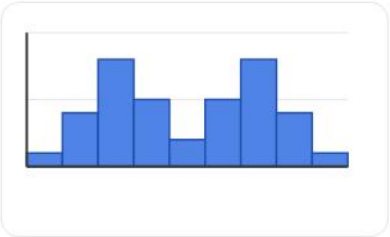
long **upper** tail, to the **right**

NEGATIVELY SKEWED



long **lower** tail, to the **left**

BIMODAL



two peaks: often two groups mixed

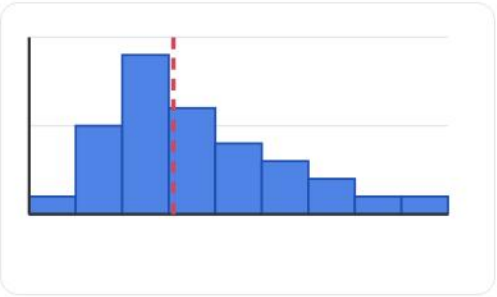
The tail names the skew. Salaries and house prices: positive. Age at death: negative.

Class discussion

LEARN · 1C

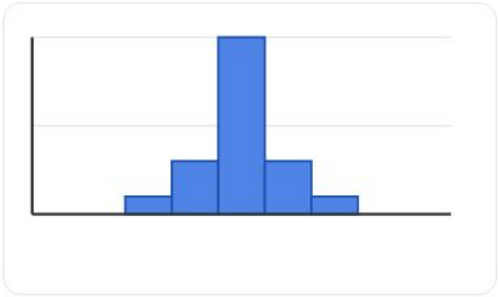
Centre, spread and outliers

CENTRE



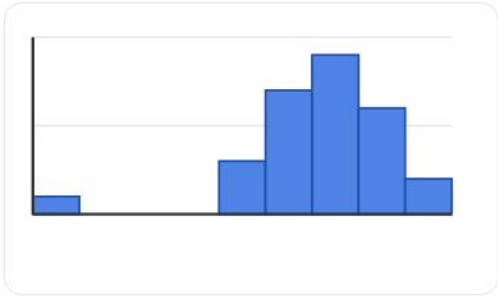
The middle: the line that splits the **area** in half.

SPREAD



Narrow peak: values tightly clustered. Broad peak: widely spread.

OUTLIERS



Values that stand out from the main body: check them, they may be errors.

From a histogram you can only **estimate** the centre and spread, and only flag **possible** outliers.

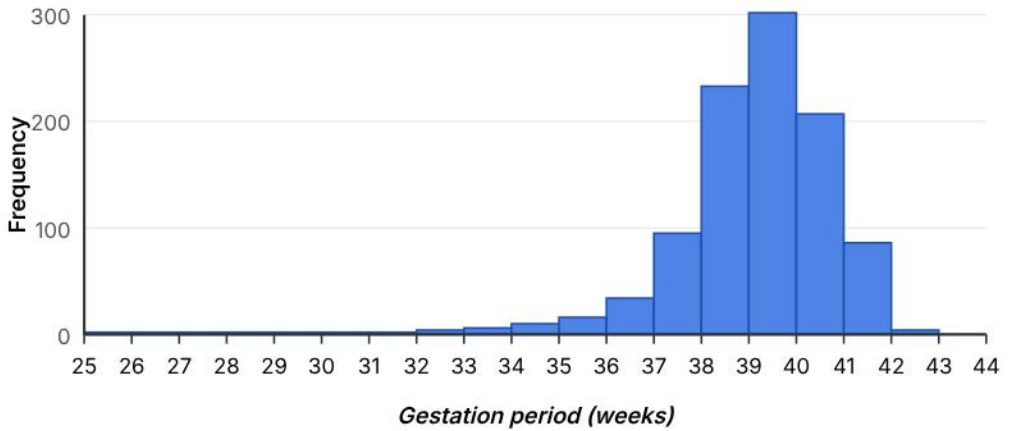
Class discussion

WORKED EXAMPLE 9 · 1C

Video

Describing the features of a distribution from a histogram

The histogram shows the gestation period (completed weeks) for a sample for 1000 babies born in Australia one year. Describe this histogram in terms of shape, centre, spread and outliers.



1 The distribution is clearly negatively skewed, with a long lower tail. 2 The centre of the distribution is around 38-39 weeks. 3 The data values range from 25-42 weeks, but most of the data values are close to the centre, in the range of 36-41 weeks. 4 Those values which are less than 34 weeks seem to be small in comparison to the rest of the data, but which values are outliers cannot be determined from the histogram.

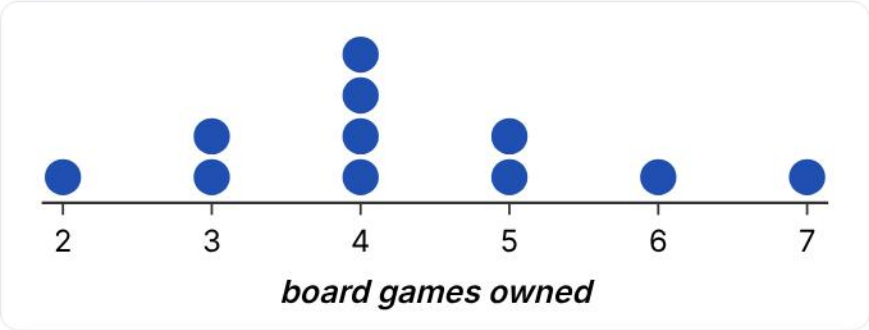
Cold calling

LEARN · 1D

Dot plots and stem plots

DOT PLOT

A number line with each data point marked by a dot. Equal values stack.



Best for small, discrete data sets.

STEM PLOT

Each value splits into a **stem** (leading digits) and a **leaf** (last digit).

Key: 4 | 3 = 43

4	3 5 5 5 7
5	1 3 6 8
6	7 9
7	1

- always add a **key**
- leaves in order
- include empty stems
- bunched? **split the stems**

Class discussion

WORKED EXAMPLE 10 · 1D

Video

Constructing a dot plot

The ages (in years) of the 13 members of a cricket team are:

22 19 18 19 23 25 22 29 18 22 23 24 22

Construct a dot plot.



1 **Number line**
Scale it to include every data value (17 to 30). Label it with the variable.

2 **Plot**
Mark a dot above each value. Equal values stack.

Cold calling

YOU TRY · 1D

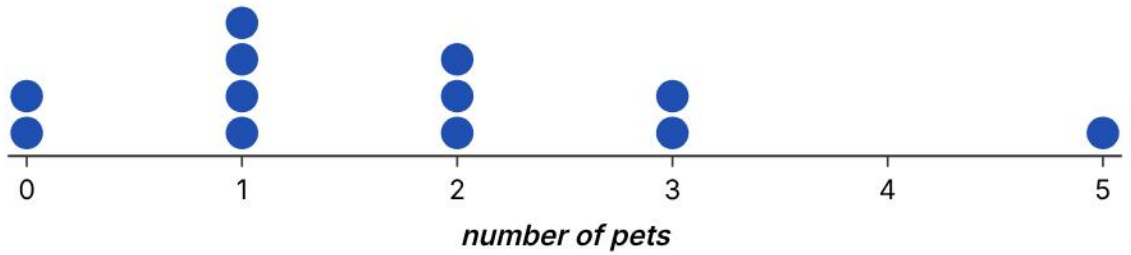
Constructing a dot plot

Sophie surveyed 12 of the families living on her street. She asked for the number of pets each of them owned and the results were recorded.

3 2 0 1 1 3 0 5 2 1 1 2

a. Construct a dot plot to display this data.

b. What was the modal *number of pets* owned by families that Sophie surveyed?



a Order: 0 0 1 1 1 1 2 2 2 3 3 5.
Scale 0 to 5; a dot for each value.

b **1 pet** (the most dots)

Quick write

WORKED EXAMPLE 11 · 1D

▶ Video

Constructing a stem plot

University participation rates (%) in 23 countries are given below.

26 3 12 20 36 1 25 26 13 9 26 27
15 21 7 8 22 3 37 17 55 30 1

Display the data in the form of a stem plot.

Unordered

0	3 1 9 7 8 3 1
1	2 3 5 7
2	6 0 5 6 6 7 1 2
3	6 7 0
4	
5	5

Ordered

rate (%) Key: 1|2 = 12

0	1 1 3 3 7 8 9
1	2 3 5 7
2	0 1 2 5 6 6 6 7
3	0 6 7
4	
5	5

1 **Stems**
Values in the units to fifties: stems 0, 1, 2, 3, 4, 5.

▶

2 **Leaves**
Work through the data: 26 → stem 2, leaf 6.

▶

3 **Order + key**
Leaves in ascending order; add the variable name and a key.

Cold calling

YOU TRY · 1D

Constructing a stem plot

Ms Smyth's maths class of 25 students sat their end-of-year exam. Their results (%) were recorded.

55 68 76 90 83 89 75 66 59 84 48 62 58
95 80 77 61 92 99 63 84 65 70 81 96

a. Construct a stem plot to display this data.

b. How many students scored above 70% on the exam?

Key: 4 | 8 = 48%

4	8
5	5 8 9
6	1 2 3 5 6 8
7	0 5 6 7
8	0 1 3 4 4 9
9	0 2 5 6 9

b 7 | 5 6 7, and every leaf on stems 8 and 9:
14 students

Quick write

LEARN · 1E

Logs are just powers of 10

number	0.01	0.1	1	10	100	1000	1 000 000
power of 10	10^{-2}	10^{-1}	10^{0}	10^{1}	10^{2}	10^{3}	10^{6}
log	**−2**	**−1**	**0**	**1**	**2**	**3**	**6**

$\log_{10} x$

If $\log_{10} x = b$, then $10^{b} = x$

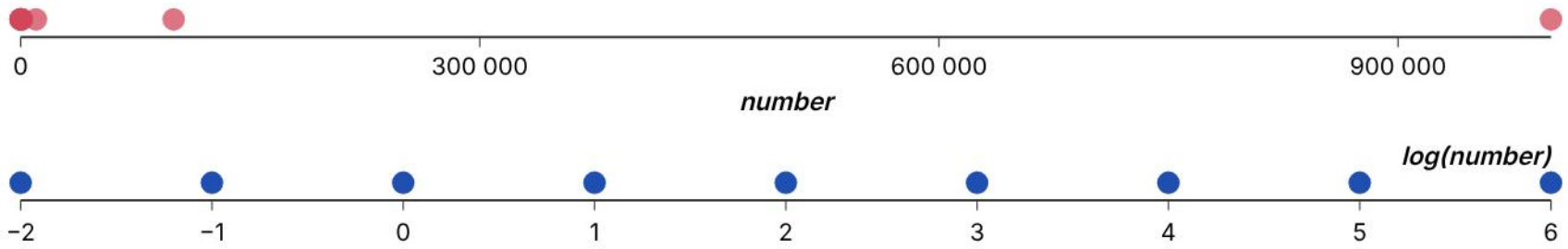
PROPERTIES

- number > 1 → log is positive
- number = 1 → log is 0
- between 0 and 1 → log is negative
- 0 or negative → log is undefined

LEARN · 1E

Why use a log scale?

0.01, 0.1, 1, 10, 100, 1000, 10 000, 100 000, 1 000 000 on an ordinary scale, then on a log scale:



ORDINARY SCALE

Eight numbers bunch at 0; one million sits alone.

LOG SCALE

Evenly spread. Use it when data range over **several orders of magnitude**. A positively skewed histogram often becomes symmetric.

Class discussion

WORKED EXAMPLE 12 · 1E

Video

Using a CAS calculator to find logs

a Find the log of 45, correct to two significant figures.

b Find the number with log equal to 2.7125, correct to the nearest whole number.

a Calculator screen: type log(45). $\log 45 = 1.65\ldots = 1.7$ (to 2 sig. figs)

b If the log of a number is 2.7125, the number is $10^{2.7125}$. $10^{2.7125} = 515.82\ldots = \mathbf{516}$ (to the nearest whole number)

Undo a log with a power of 10: type 10^2.7125 on the calculator screen.

Cold calling

YOU TRY · 1E

Using a CAS calculator to find logs

Use $\log_{10}$ for the following calculations.

a. Calculate the log of 23, correct to two decimal places.

b. If the log of a number is 5, what is the number?

a $\log_{10}(23) = 1.361\ldots \rightarrow$ **1.36**

b $\log_{10}(x) = 5$ so $x = 10^5 \rightarrow$ **100 000**

Quick write

WORKED EXAMPLE 13 · 1E

Video

Interpreting a histogram with a log scale

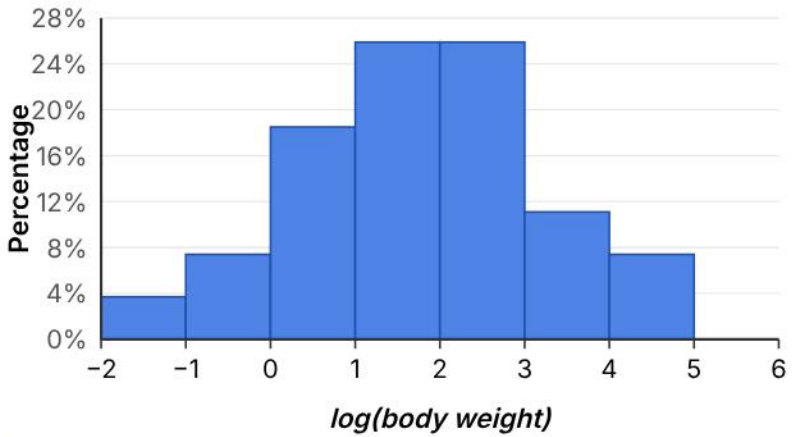
The histogram shows the distribution of the weights of 27 animal species plotted on a log scale.

a What body weight (in kg) is represented by the number 4 on the log scale?

b How many of these animals have body weights more than 10 000 kg?

c The weight of a cat is 3.3 kg. Use your calculator to determine the log of its weight correct to two significant figures.

d Determine the weight (in kg) of the animal with a log(body weight) of 3.4 (the elephant). Write your answer correct to the nearest whole number.



a $10^4 = 10\,000$ kg

b 10 000 is 4 on the log scale; bars above 4: **two animals**

c $\log 3.3 = 0.518\ldots = 0.52$

d $10^{3.4} = 2511.88\ldots = 2512$ kg

Cold calling

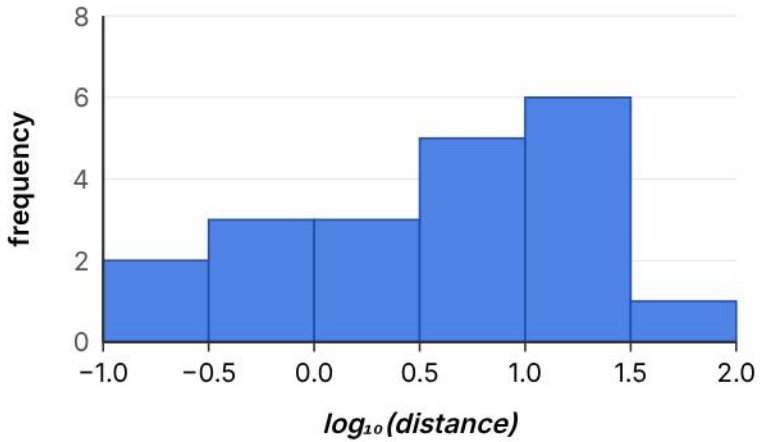
YOU TRY · 1E

Interpreting a histogram with a log scale

A class of twenty students were asked about the *distance*, in kilometres, from their home to the school. The results were summarised in a histogram using a $\log_{10}$ scale.

a. Jane was away the day of the survey. It is known that she lives 9.5 km away from school. Which interval on the histogram reflects this?

b. How many students live more than 10 km away from school?



a $\log_{10}(9.5) = 0.9777\ldots \rightarrow$ **0.5–<1.0**

b $\log_{10}(10) = 1$; columns above 1: $6 + 1$ = **7 students**

Quick write

EXIT TASK 1 OF 3 · VCAA 2019 EXAM 1 Q1

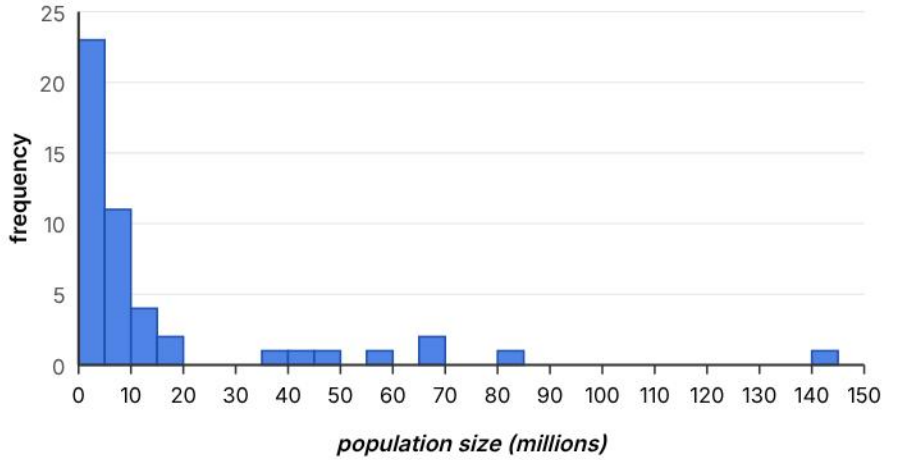
VCAA: **92%** of students answered correctly

Show what you know

The following histogram shows the distribution of the *population size* of 48 countries in 2018.

The number of these countries with a *population size* between 5 million and 20 million is

A. 11	3%	**B.** 17	92%	**C.** 23	2%
D. 34	1%	**E.** 35	2%		



SOLUTION · B

The 5–10, 10–15 and 15–20 columns: $11 + 4 + 2 = 17$

WHAT'S THE DISTRACTOR?

No clear distractor: only 8% chose a wrong answer.

Multiple choice

Quick write

EXIT TASK 2 OF 3 · VCAA 2016 EXAM 1 Q3

VCAA: **87%** of students answered correctly

Show what you know

The following stem plot displays 30 temperatures recorded at a weather station.

The modal *temperature* is

A.	**2.8 °C**	**87%**
B.	2.9 °C	9%
C.	3.7 °C	1%
D.	8.0 °C	2%
E.	9.0 °C	1%

temperature Key: 2 | 2 = 2.2 °C

2	2 2 4 4
2	5 7 8 8 8 8 8 8 9 9 9 9
3	1 2 3 3 4 4 4
3	5 6 7 7 7 7
4	1

SOLUTION · A

The leaf 8 appears six times on the 2 stem: 2.8 °C. (The key: 2 | 2 = 2.2.)

WHAT'S THE DISTRACTOR?

No clear distractor. 9% chose B: 2 | 9 has four leaves, but 2 | 8 has six.

EXIT TASK 3 OF 3 · VCAA 2019 EXAM 1 Q3

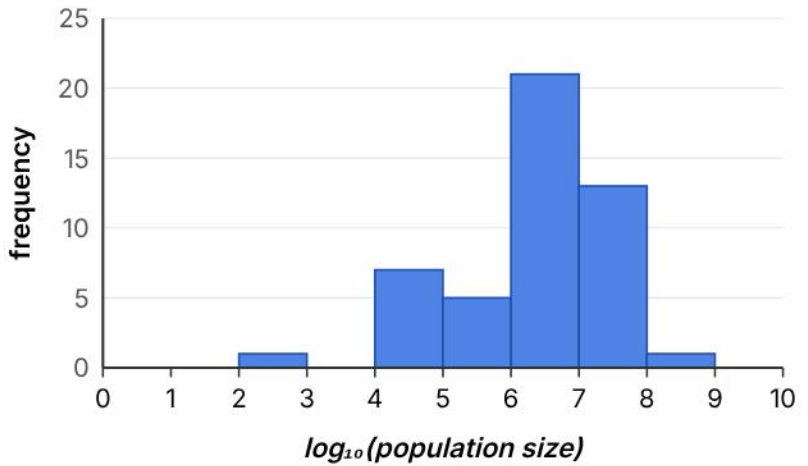
VCAA: **71%** of students answered correctly

Show what you know

The following histogram shows the *population size* for 48 countries plotted on a $\log_{10}$ scale.

Based on this histogram, the number of countries with a *population size* that is less than 100 000 people is

A. 1	7%	**B.** 5	6%	**C.** 7	7%
D. 8	71%	**E.** 48	9%		



Data: Worldometers

SOLUTION · D

$100\,000 = 10^5$, so 5 on the log scale. Columns below 5: $1 + 7 = 8$

WHAT'S THE DISTRACTOR?

No clear distractor: wrong answers were spread (6–9% each).

Multiple choice

Quick write

CHECK-IN

How did you go?

- ☐ I can construct a frequency table for discrete numerical data.
- ☐ I can construct a grouped frequency table.
- ☐ I can construct a histogram from a grouped frequency table.
- ☐ I can describe the features of a distribution from a histogram.
- ☐ I can construct a dot plot.
- ☐ I can construct a stem plot.
- ☐ I can use a CAS calculator to find logs.
- ☐ I can interpret a histogram with a log scale.

Thumbs up, sideways or down for each one, then rate it on your worksheet: ✓ got it · ~ nearly · × not yet



True or false

HOMEWORK

Homework

CAMBRIDGE · EXERCISE 1C

Frequency table, discrete data	**Q1–2**
Grouped frequency table	**Q3**
Histogram from a frequency table	**Q4**
Reading a histogram	**Q5–6**
Histogram on CAS	**Q7–8**
Shape, centre and spread	**Q9–10**
Exam 1 style questions	**Q11–12**

CAMBRIDGE · EXERCISE 1D

Dot plots	**Q1–5**
Stem plots	**Q6**
Split stems	**Q7**
Interpreting stem plots	**Q8–9**
Exam 1 style questions	**Q10–11**

CAMBRIDGE · EXERCISE 1E

Log of a number	**Q1**
Number from its log	**Q2**
Histogram with a log scale	**Q3**
Interpreting a log scale	**Q4**
Exam 1 style questions	**Q5–6**

REVISION BOOK

The **1C**, **1D** and **1E** sections